

Configure IPv6 Addressing

Task – 1 Command line in windows to assign the IP address.

- Microsoft Windows is a widely used graphical operating system family developed by Microsoft, powering personal computers, enterprise servers, and embedded systems. Graphical User Interface (GUI): Navigates file paths and applications visually using a desktop, taskbar, icons, and a Start menu. Multitasking & Hardware Support: Runs multiple applications concurrently while supporting plug-and-play hardware devices. Central File System: Uses NTFS for file management, directory security, and drive encryption via tools like BitLocker. System Management: Configurations are managed centrally through the Control Panel, Settings app, File Explorer, and PowerShell / Command Prompt.
- Windows 10 / 11: Shifted toward a unified interface, security integration, regular feature updates, and built-in cloud.

Task – 2 Checking the connectivity of the remote desktop.

- Cmd – write the ping and the IP address.

Task – 3 Things are changing after giving temporary IP address.

- To set up and observe a temporary IPv6 address in Windows, enable privacy extensions using `netsh interface ipv6 set privacy state=enabled`, run `ipconfig /renew6` to generate a temporary address, and verify its status using `ipconfig /all`. When you disconnect and reconnect your network interface, the temporary IPv6 address typically remains unchanged as long as its preferred lifetime timer has not expired; however, upon restarting Windows, the system clears volatile RAM and generates a completely new randomized temporary IPv6 address to prevent tracking across sessions while keeping your primary IPv6 address constant.

Task – 4 Connectivity with partial name on scenario

- Script using the coding language

Task – 5 Subnetting the scenario.

- 1.Open Command Prompt or PowerShell as Administrator :-Press Win + X and select Terminal (Admin) or PowerShell (Admin), or search for cmd in the Start menu, right-click it, and select Run as administrator.
- 2.Select Your IPv6 Subnet & Address :- From the AI-generated scenario using the base prefix 2001:db8:abcd::/48, select the Engineering Subnet (2001:db8:abcd:1::/64) and choose the host IP 2001:db8:abcd:1::50.
- 3.Assign the IPv6 Address via Command Line :- Run the following command (replace "Ethernet" with "Wi-Fi" if connected wirelessly):

```
- DOS
- netsh interface ipv6 add address interface="Ethernet"
  address=2001:db8:abcd:1::50/64
```

- 4.Verify System Configuration :- Confirm that Windows has bound the address to your network adapter. Verification: Under your active adapter, confirm 2001:db8:abcd:1::50 is listed under IPv6 Address.

```
- DOS
- ipconfig
```

- 5.Test Self-Connectivity :- Test that your system responds on the newly configured IPv6 address. Verification: You should receive direct replies (Reply from 2001:db8:abcd:1::50: time<1ms) with 0% packet loss.

```
- DOS
- ping -6 2001:db8:abcd:1::50
```

- Removing the Test Address
- When you finish testing, remove the manually assigned static IPv6 address by running:

```
- netsh interface ipv6 delete address interface="Ethernet"  
address=2001:db8:abcd:1::50
```